Software Requirement System
for
Home Security System

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1.1 Purpose

The purpose of this document is to outline the functional and non-functional requirements of the Home Security System. It serves as a guideline for system development and implementation, ensuring that the final product meets the needs and expectations of users.

1.2 Scope

The Home Security System includes the following features:
Access Control: Defining who can enter the home and how, such as through keypads, keycards, or biometric systems.
Intrusion Detection: Detect unauthorized entry into the premises through door and window sensors.
Motion Detection: Detect motion within the premises using motion sensors.
Video Surveillance: Monitor the surroundings through connected security cameras.

Alarm System: Trigger audible and visual alarms in response to security breaches.
Remote Monitoring: Allow users to monitor the security system remotely via a mobile application or web interface.
Emergency Response: Notify homeowners and authorities in the event of a security breach or emergency.
Backup Systems: Implementing backup power sources and communication methods to ensure the system remains operational during power outages or network failures.

1.3 Definition, Acronym, and Abbreviation

Java – platform independence
SQL – Structured Query Language
DFD – Data Flow Diagram
CFD – Context Flow Diagram
ER – Entity Relationship
IDE – Integrated Development Environment
SRS – Software Requirement Specification

2. Overall Description

2.1 Product Perspective

From a product perspective, a home security system by SRS should aim to provide comprehensive protection, ease of use, and integration with modern technology. Here's a breakdown:
Hardware: The system should include a variety of hardware components such as cameras, motion sensors, door/window sensors, smart locks, and a central control panel. These devices should be of high quality, durable, and aesthetically pleasing to fit into the home environment seamlessly.
Software: A user-friendly mobile app and web interface are essential components. Users should be able to view live video feeds, receive real-time alerts, and various aspects of the system remotely.

Security Features: The system should offer advanced security features such as encryption to protect user data, tamper detection on devices, and backup options in case of power outages or internet failures. Additionally, it should provide options for professional monitoring services and emergency response integration for added peace of mind.
Thus, in addition to security, there is also integration, customization, scalability, and customer service.

2.2 User Class and Characteristics

Homeowner: All of the security functions of the system can be performed by the homeowner who has the control panel password.
Web user: All of the system's functions can be performed by the web user, locally in the Wireless 6 house or remotely from afar, who has the Web user ID and password as well as the control panel password.
Monitoring company: The monitoring company can receive the alarm from the user, the sensors, or motion detectors as well as respond to alarming events.
Sensors and motion detectors: The sensors and motion detectors can invoke the system's alarming functions in the case of intruder or accident detection.

2.3 Operating Environment

The operating Environment part of a SRS document outlines the circumstances under which the software will function. For home security software, this section would encompass various

aspects such as:

Hardware Requirements: Detail the hardware necessary for the software to run effectively. This could include specifications for cameras, sensors, control panels, and any other devices integrated into the home security system.

Software Requirements: Outline the software dependencies of the home security system. This might include operating systems, databases, communication protocols, and any other software components necessary for the system to function.

Network Requirements: Specify the network environment needed for the software to operate, such as internet connectivity requirements, network protocols, bandwidth considerations, and any security measures needed for data transmission.

Physical Environment: Describe the physical conditions in which the software will be deployed. This could involve considerations like temperature, humidity, power supply, and physical security measures for the hardware components.

Integration with Existing Systems: If the home security software needs to integrate with other systems or devices (such as smart home platforms or third-party security services), detail the compatibility requirements and any protocols or APIs that need to be supported.

User Environment: Describe the environment from the user's perspective, including the devices they will interact with (e.g., smartphones, tablets, desktop computers), supported operating systems, and any user interface requirements.

Regulatory and Compliance Requirements: Address any legal or regulatory requirements that impact the operating environment.
Environmental Considerations: Consider any environmental factors that could affect the performance or reliability of the home security system, such as weather conditions or potential sources of interference.

2.4 Design and Implementation Constraints

Budget Constraints: The cost of the system should be within the budget constraints of the homeowner.
Hardware Limitations: The choice of hardware components may be limited by factors such as the size of the home, power supply availability, and compatibility with existing infrastructure.
Scalability: The system should be scalable to accommodate future expansions or upgrades. This includes adding additional sensors, cameras, or smart devices as needed.
Power Consumption: Minimizing power consumption can prolong battery life and reduce the need for frequent replacements.
Compatibility: The system should be compatible with a variety of devices and platforms commonly used in homes.
Reliability and Availability: The system should be highly reliable and available at all times to ensure continuous protection.
Data Privacy and Security: The system should adhere to

industry standards and best practices for encryption, authentication, and access control.
detectors, etc.

Regulatory Compliance: Compliance with relevant laws and regulations. This includes regulations regarding the use of surveillance cameras and the storage of personal data.
User Interface Constraints: The user interface should be intuitive and easy to use, catering to users of varying technical expertise.
Installation and Maintenance: The system should be easy to install and maintain, with clear instructions provided to users.
Co1 – UML 2.0 shall be used for the system's analysis and design.
Co2 – Java shall be used as the programming language to be compatible with the provided devices' interfaces.

2.5 User Documentation

UD-1: The system shall provide an online hierarchical and cross-linked help system in HTML that describes and illustrates all system functions.
UD-2: The system shall provide an offline help system in CHM that describes and illustrates all system functions.
UD-3: The project shall provide a soft copy of the user manual for the homeowners who buy the system.
UD-4: The project shall provide the user with the manuals of the hardware devices such as cameras, sensors, motion detectors, etc.

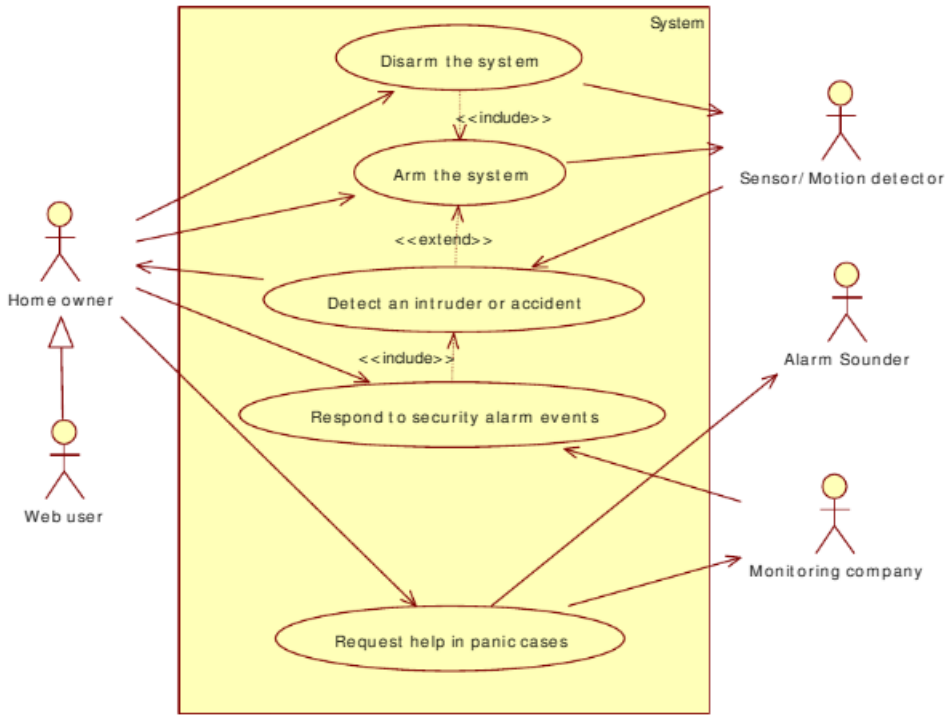
3. System Features

3.1 Secure the house

3.1.1 Descriptions

The homeowner whose identity has been verified can arm/disarm the house, using the control panel or Web interface, with the door sensors, window sensors, motion detectors, fire, CO, smoke level sensors, and basement water level sensors. If an intruder or an accident is detected, the system shall alarm and inform the monitoring company. Homeowners themselves can announce emergency cases (such as heart attacks) by requesting the system to sound the alarm and call the police. In addition, these controls can be performed either locally or via remote devices such as mobile phones or PCs using a Web interface.

3.1.2 Use Case Diagram



3.1.3 Functional Requirements

Requirement	Descriptions
Secure.Operation	The system shall operate this feature's functionalities independently with the "Observe the house" feature, but exclusively with the "Configure the system" feature.
Secure.Operation.Interface	This feature can be performed via either control panel or web interface, supported for both local network and remote control.
Secure.Operation.Number	The system shall allow multiple control panel interfaces, whereas only 1 web user can use this feature at a time.
Secure.Operation.Display	This system shall display its current status and if there is some task in operation.
Secure.Operation.Ready	When powered or disarmed, the system shall check if the connection status with the sensor-attached devices and cameras is ready.
Secure.Check	The system shall check the house condition whenever a sensor or detector activating/deactivating task is requested.
Secure.Check.Error.Repeat	If a house error condition is met, periodically, the system shall automatically check the house condition until all errors are resolved.
Secure.Check.Error.Inform	System shall clearly inform the user what condition error is. In addition, only one error is displayed in control panel, whereas all of the errors can be displayed via web interface
Secure.Password	With the control panel interface, a 4-digit password will be asked. With the Web interface, there are 2 levels of passwords, the web user id and password, and the control panel password.

Secure.Password.ControlPanel	The system shall ask the password for every request from the control panels, except for panic function
Secure.Password.WebInterface	The system shall ask the password only once. Tasks can be performed without any password request again. The session time-out is set to be 10 minutes.
Secure.Password.CPWrong	With the control panel password, the system shall allow only five continuous wrong passwords in 15 minutes. If the condition is broken, the system shall alarm and inform the monitoring company.
Secure.Password.WIWrong	With the web interface password, the system shall allow only five continuous wrong passwords. The session will be blocked for 15 minutes if this condition is broken, and the failure log will be saved to server as well as sent to the monitoring company.
Secure.Password.Store	In the case of web interface, the system shall only allow to save the web user id and password to the browser's cookie. The control panel password shall be asked for every log-in.
Secure.Mode	The system shall support two standard modes: Stay mode for home owner staying at home and Away mode for home owner who is going to go out.
Secure.Mode.ControlPanel	In the case of control panel access, the system shall only allow the home owner to arm/disarm the sensors by the Stay/Away modes.
Secure.Mode.WebInterface	In the case of web interface access, the system shall allow the home owner to arm/disarm the sensors according to the predefined security zones as well as arm/disarm the sensors individually.
Secure.Mode.Change	In the case of control panel access, the system shall ask the password whenever the user wants to change the modes.
Secure.Mode.Disarm	In the case of control panel access, the system shall ask the control panel password before allowing home owner to disarm and let the system at the waiting (ready) state.

4. External Interface Requirements

4.1 User Interface

1. The system shall provide a user-friendly interface accessible via a web browser or mobile application.
2. The UI shall allow users to arm, disarm, and configure the security system distantly.
3. It shall display real-time status updates, such as sensor triggers, alarms, and camera feeds.

4.2 Sensor Interface

1. The system shall support various types of sensors, including motion sensors, door/window sensors, and glass break detectors.
2. It shall interface with these sensors to monitor the status of entry points and detect unauthorized access or intrusion.
3. It shall provide a means to configure sensor settings and sensitivity levels.

4.3 Alarm Notification Interface

1. The system shall notify users via email, SMS, or push notifications in the event of an alarm trigger.
2. It shall provide options for users to customize notification preferences and recipient contacts.

4.4 Camera Interface

1. The system shall integrate with surveillance cameras for live

monitoring and recording.
2. It shall support standard camera protocols.
3. Users shall be able to view camera feeds through the system's interface and receive alerts based on camera activity.

4.5 Remote Access Interface

1. The system shall provide secure remote access for users to control and monitor their security system from anywhere with internet connectivity.
2. It shall implement robust authentication mechanisms.

5. Other Nonfunctional Requirements

5.1 Performance requirements

Response Time: The system should respond to user inputs within an acceptable timeframe.
Scalability: The system should be able to handle an increasing number of users and devices without significant performance degradation.

5.2 Security Requirements

Data Encryption: All communication between devices and servers should be encrypted to prevent unauthorized access.
Access Control: Users should have different levels of access based on roles.
Intrusion Detection: The system should be able to detect unauthorized access attempts and notify the user or authorities.

5.3 Regulatory Compliance

Compliance with Data Protection Regulations: The system should comply with relevant data protection laws and regulations, such as GDPR or CCPA.
Safety Regulations: The system should adhere to safety regulations to prevent any hazards to users or their property.

5.4 Software Quality Attributes

SQA – 1: User can request service for any hardware/software errors and malfunctions at any time. The company will respond by telephone or visit the customer to fix the customer's request when the request is received.
SQA – 2: Control panels are installed anywhere inside the house, but one shall be installed in front of the main door since the user will have enough time when leaving and getting back to the house.
SQA – 3: The web interface shall be easy to use and provide as convenient and easy to use as the control panel provides.
SQA – 4: When the Internet connection is broken, the system shall ask the homeowner for changing to the telephone line.
SQA – 5: Regardless of the user's IT capability, drawing of the house's floor plan shall be serviced.

Appendix: Glossary

Home security: the activities to maintain the safety of the house such as protecting the house from unauthorized intruders, restricting the house access from authorized visitors, and detecting unexpected dangers (fire) or accidents (heart attack).

Home surveillance: the activities to keep track of the house's status via the cameras placed inside and outside.

Web interface: the user system's interaction interface that accepts input and provides output by generating web pages which are transmitted via the (Internet or local) network and viewed by the user using a web browser program.

Security zone: Some of the sensors and motion detectors are grouped to define a security zone. The security zones regulate the authority and restriction of parts of the house and may overlap each other.

Surveillance zone: Some of the cameras are grouped to define a surveillance zone. The surveillance zones regulate the homeowner's observation of parts of the house and may overlap each other.

"Stay"/"Away" modes: The "Stay"/"Away" modes are the system's configurations in which some of the security zones are enabled and some of the surveillance zones are enabled. Literally, the "Stay" mode provides barely enough security and surveillance to comfort the staying homeowner, while the

“Away” mode provides a standard high level of security and surveillance around the house when the homeowner is away.

Sensor: A device that measures a physical quantity and converts it into a signal that can be read by an observer or by an instrument.

Motion detector: A device that contains a physical mechanism or electronic sensor that quantifies motion that can be either integrated with or connected to other devices that alert the user of the presence of a moving object within the field of view.

Arm/disarm: The activities to activate/deactivate some predefined sensors and motion detectors in/outside of the house. Arm has an abstract meaning which can be set as “Stay” mode, “Away” mode, or a complex mode. Whereas, disarm has a single meaning of deactivating all the in-operation sensors and motion detectors.

House condition: The condition of the devices or parts of the houses that are sensor-attached. An opened door or window may cause the house condition an error condition when the homeowner wants the system to arm the house.

Central processor: The central processor encompasses the control software, which can be accessed via a control panel interface or web interface, and a web server to support the wireless network over the house as well as the remote control far away from the house.